

# IRFAN ULLAH

## SENIOR MOBILE APPLICATION, AI & GAME ENGINEER

Innovative Software Engineer with 2+ years of production experience crafting robust cross-platform Flutter applications, casual 2D mobile games, and generative AI systems. Specialized in developing multimodal voice assistants (Gemini, LiveKit, Vapi) and computer vision pipelines (image synthesis, face swap, neural video editors). Proven mastery in full lifecycle deployment across Google Play Store, Apple App Store, and production Linux VPS web infrastructure.

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### WORK EXPERIENCE

#### Senior Flutter & Mobile Engineer — Socioon

12/2023 – Present

Lahore, Pakistan

- Architecting core social media and generative AI components, driving user retention and modular scalability.
- Engineered high-concurrency Agora audio/video calling pipelines, live stream broadcasting, interactive storytelling, and secure peer-to-peer digital wallets.
- Integrated multi-tier payment gateways (Stripe & PayPal) along with Google AdMob revenue streams and OneSignal behavioral push notification logic.
- Enforced robust state management architectures using BLoC, Riverpod, and GetX with decoupled data repositories.

#### Flutter Developer Intern — Heapware

03/2022 – 12/2022

Lahore, Pakistan

- Developed cross-platform Android and iOS productivity utilities, task reminders, and coffee ordering commerce applications.
- Implemented reactive state handling via GetX and Provider, optimizing async REST API synchronization and responsive user interfaces.

### PRODUCTION SMART VOICE ASSISTANTS

#### Nexivo AI Voice Assistant

Completed & Live

##### Autonomous Voice Agent ([nexivoai.co/ai-assistant](https://nexivoai.co/ai-assistant))

- Engineered an ultra-low latency voice-first assistant integrating Vapi's conversational speech pipeline with sub-second response times.
- Configured custom LLM reasoning workflows, speech-to-text transcription, voice synthesis, and dynamic webhook trigger automation.

#### LiveKit Real-Time Multimodal Voice Agent

Active Engineering

##### Ultra-Low-Latency WebRTC Assistant

- Building bidirectional audio streaming with LiveKit WebRTC transport protocols enabling instant human-like speech interruption (barge-in).
- Integrated Voice Activity Detection (VAD) and noise suppression for high-fidelity cross-platform voice experiences.

#### Gemini Multimodal AI Assistant

Active Project

##### Google Gemini 1.5 Pro & Flash Mobile Pipeline

- Developing real-time multimodal assistance supporting streaming Markdown responses, document analysis, and visual image inspection.

### TECHNICAL SKILLS

#### Mobile & Game Engines

Flutter   Dart   2D Game Engine   BLoC  
Riverpod   GetX   Clean Architecture

#### AI & Voice Systems

Gemini Multimodal   Vapi AI  
LiveKit WebRTC   ElevenLabs TTS  
AI Image Gen   Neural FaceSwap  
AI Video Editing

#### DevOps, Store & Cloud

Google Play Console   Apple App Store  
Linux VPS (Ubuntu)   Nginx & SSL  
Firebase   Agora RTC   Stripe & PayPal  
Git & GitHub

### EDUCATION

#### BS in Computer Science

##### Virtual University of Pakistan

10/2018 – 03/2023

Core studies: Data Structures & Algorithms, OOP, Database Management Systems (DBMS), and Operating Systems.

#### ICS (Computer Science)

##### Govt. Islamia College Civil Lines, Lahore

05/2016 – 04/2018

### CERTIFICATIONS

#### Mobile App Development

##### NAVTTTC, Lahore

Intensive certification in professional cross-platform Flutter application engineering and modern mobile UI design patterns.

### LANGUAGES

English   Professional Working  
Urdu   Native / Bilingual

MOBILE GAME DEVELOPMENT

**Nedo Bird (Casual Arcade Game)** Published on Google Play  
[play.google.com/store/apps/details?id=com.bathan.nedobird](https://play.google.com/store/apps/details?id=com.bathan.nedobird)

- Designed and developed an engaging 2D tap-to-fly arcade game with real-time physics, custom sprite animations, and precise collision detection.
- Optimized frame render loops ensuring consistent 60 FPS performance across budget and flagship mobile devices.
- Integrated local high-score persistence, audio SFX feedback, and Google AdMob banners/interstitial monetization.

**Casual 2D Mini-Game (New Title)** Google Play Closed Testing  
**In-Flight Production Project**

- Building dynamic level-progression mechanics, particle effects, and responsive touch controls for casual gamers.
- Actively managing release tracks via Google Play Console internal testing to gather crash telemetry and user feedback.

GENERATIVE AI MEDIA APPLICATIONS

**AI Media Studio (Image Gen, Editor, FaceSwap & Video)** Active Engineering  
**Neural Diffusion & Computer Vision Mobile Suite**

- **AI Image Generator:** Built intuitive prompt-based image synthesis mobile client with custom aspect ratios, style seeds, and negative prompting.
- **AI Photo Editor & Inpainting:** Enabled real-time interactive canvas masking, smart object erasure, and high-resolution upscaling.
- **Neural FaceSwap Engine:** Implemented seamless face replacement pipelines with facial landmark alignment and color-matching blending.
- **AI Mobile Video Editor:** Added automated subtitle generation, audio separation, and AI visual enhancement filters.

COMMERCIAL SOCIAL & ENTERPRISE PROJECTS

**LinkOn (Paroter)** Social Networking Platform  
• Delivered live streaming broadcasting, interactive stories, audio/video calling with Agora, and in-app digital tipping wallet.

**PTI (Rabta) Mobile App** Public Outreach & Data Collection  
• Constructed responsive UI modules, secure citizen data collection forms, and high-throughput content feed distribution.

**Bandhu** Community & Social Platform  
• Created location-based friend discovery, real-time messaging, invite mechanisms, and integrated digital payment wallets.

END-TO-END DEPLOYMENT

**Google Play Store Deployment**  
Complete release management: App Bundle (AAB) builds, keystore generation, target API compliance, automated testing tracks (Internal, Closed, Open), Data Safety declarations, and store listing optimization.

**Apple App Store Deployment**  
Full publishing lifecycle: Apple Developer Account orchestration, Certificates, App IDs, Provisioning Profiles, TestFlight beta distribution, and App Store Connect review compliance.

**Web & Server VPS Deployment**  
Hands-on configuration of Linux VPS environments (Ubuntu/Debian), Nginx reverse proxy routing, Let's Encrypt SSL/TLS automation, process managers (PM2/systemd), firewall rules, and custom domain DNS mapping.

SPECIALIZED COMPETENCIES

- Conversational Voice UI Architecture
- WebRTC Bidirectional Audio Pipelines
- Casual 2D Physics & Game Loops
- State Machine Architecture (BLoC)
- In-App Purchases (IAP) & Subscriptions
- Secure Multi-Gateway Payments
- Production DevOps & Cloud VPS
- Cross-Platform Performance Profiling

DEVELOPMENT TOOLKIT

VS Code

Android Studio

Xcode

Postman

Figma

Nginx

SSH / Linux CLI

Git & GitHub

PROFESSIONAL STATUS

- Available for high-impact Remote and Onsite roles.
- Experienced in taking applications from concept through development, store review, and server hosting.